

ARIF HUSSAIN

arif.hussain1092004@gmail.com | <https://github.com/arif-hussain09>

EDUCATION

Delhi Technological University Delhi, India

BTech in Mathematics and Computer Science Graduation: Aug 2027

TECHNICAL SKILLS

Languages: Python, C

Machine Learning & Deep Learning: ANN, RNN, CNN, Transformer Architectures, Statistical Modeling & Data Analysis

Generative AI: Diffusion Models, Vision Transformers (ViTs), LangChain, LangGraph

Computer Vision: OpenCV, MediaPipe

Tools: VS Code, Google Colab, Jupyter Notebook

RESEARCH EXPERIENCE

Meity Delhi, India

Research Intern Jun 2026 – Aug 2026

- **Worked on a hybrid AI-based Intrusion Detection System (IDS)** integrating supervised learning for known attack detection and unsupervised learning for zero-day anomaly detection, improving the framework's ability to identify both existing and emerging cyber threats.
- **Researched global AI ecosystems and national AI strategies**, analyzing AI education policies, industrial adoption, government investments, industry-academia collaborations, and innovation initiatives across leading countries to identify best practices and emerging trends.

Key Project — ML/Applied-Math Approaches to Early Detection of Zero-Day IoT Botnet Propagation (Mirai-class threats)

- Built a cross-dataset IoT botnet detection framework, training on **CICIoT2023** and testing zero-day generalization on **MedBioT** (both open-source datasets).
- Applied feature selection and **Optuna-based hyperparameter tuning** across multiple ML models to optimize detection and inference performance.
- Implemented **CUSUM/EWMA-based early detection** to identify low-and-slow botnet propagation before DDoS activity.

PROJECT EXPERIENCE

Personal Projects Delhi

Jul 2023 – Present

- Developed a Movie Recommendation System using machine learning techniques.
- Built a Farm Resource Management System for efficient agricultural resource allocation.
- Implemented object detection and a hand-gesture-based volume controller using computer vision (OpenCV, MediaPipe).

LEADERSHIP EXPERIENCE

Delhi Technological University Delhi, India

MCE Students Community Jan 2024 – Present

- Elected President of the MCE Students Community, leading departmental initiatives and student engagement.
- Cultivated a community of 200+ applied mathematics students through bi-weekly workshops and networking events, increasing student engagement by 40% within the first year.

NEP SAARTHI Jan 2024 – Feb 2025

- Served as NEP SAARTHI, a student ambassador for academic reforms in transforming higher education in India.